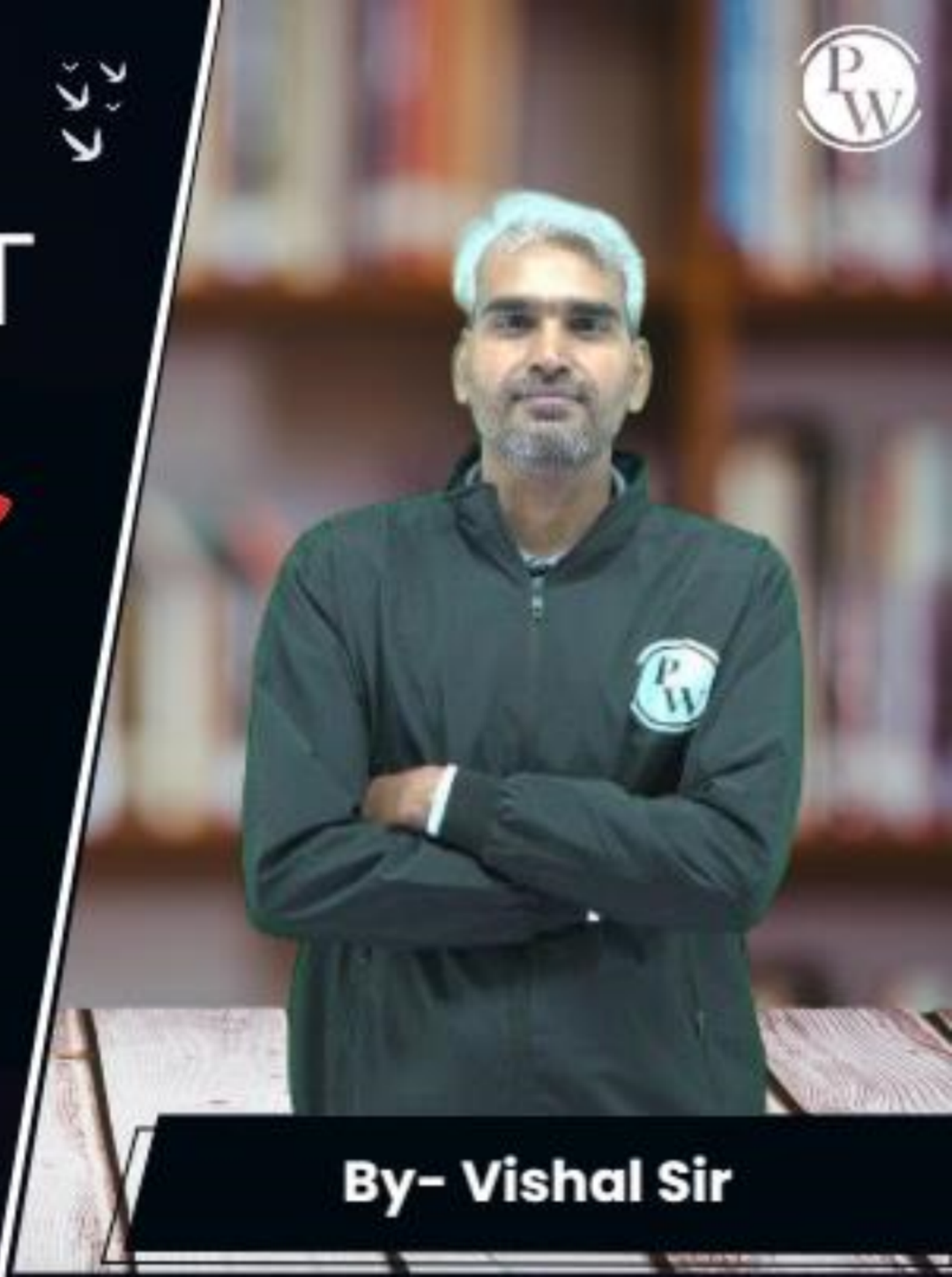
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 15



By- Vishal Sir

Recap of Previous Lecture



Topic — Lattice

Topic — Hasse diagram / POSET diagram

Slide

Topics to be Covered



Topic Bounded lattice

Topic Complement of an element in a lattice

Topic Complemented lattice

Slide

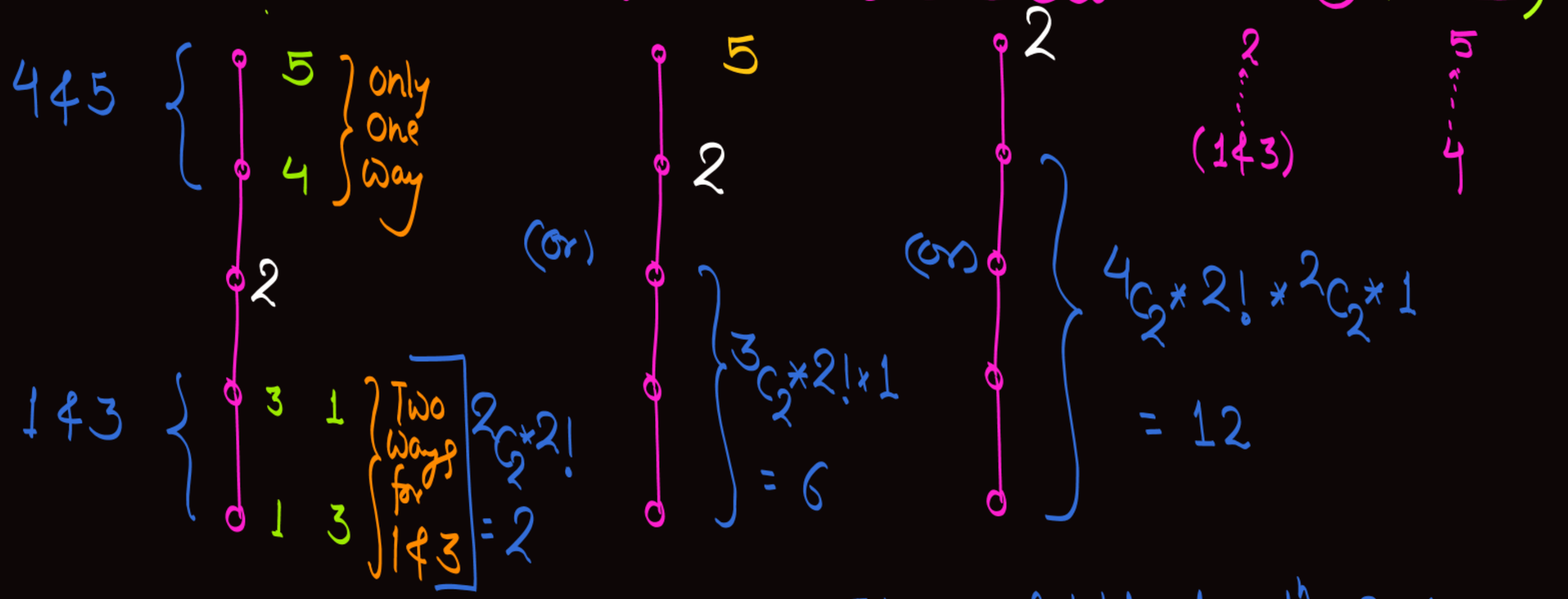
H.W. Q:- Let $A = \{1, 2, 3, 4, 5\}$

and let $R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (3,2), (4,5)\}$ is a Partial order on set A.

How many total order relation are possible on set A such that relation 'R' is included in each total order relation

H.W. Q:- Let $A = \{1, 2, 3, 4, 5\}$

and let $R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (3,2), (4,5)\}$



Total no. of total order rel^n = 2+6+12 = 20
on set A such that
Partial order rel^n R is included in each such total order rel^n.

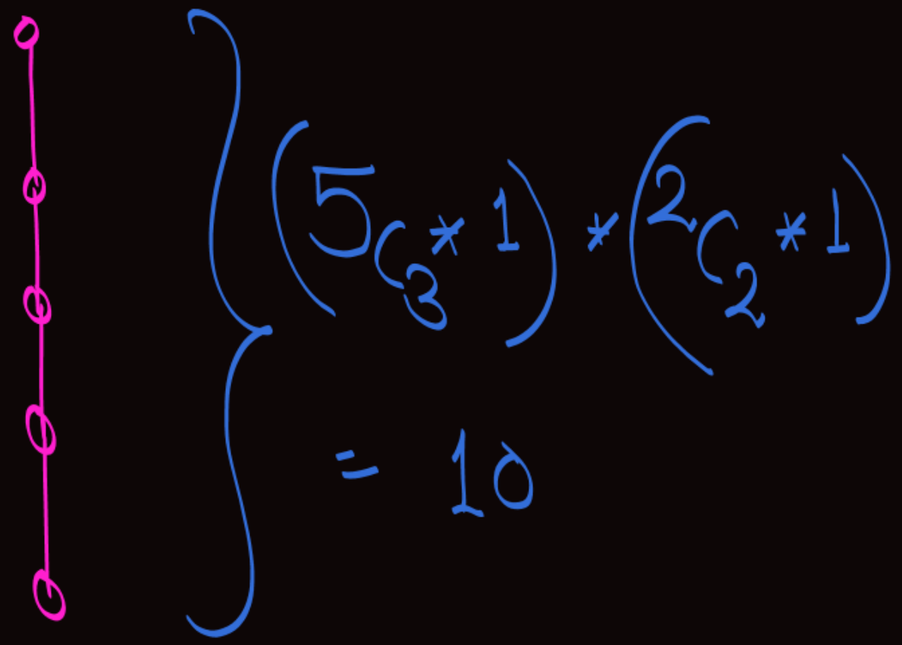
Q:- Let $A = \{1, 2, 3, 4, 5\}$

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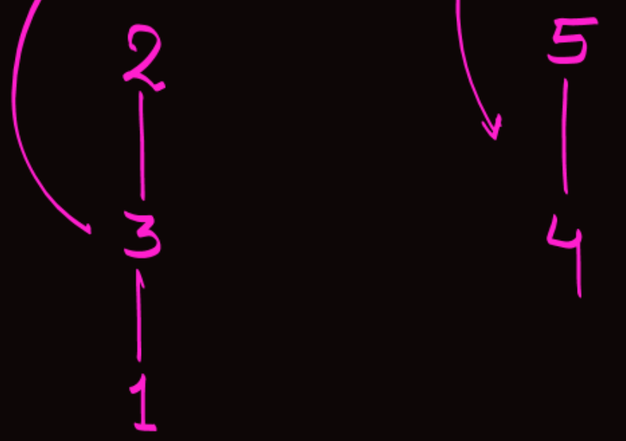
Q:- Let $A = \{1, 2, 3, 4, 5\}$

and let $R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (1,3), (3,2), (4,5)\}$ is a Partial order on set A



$$(5_{C_3} * 1) * (2_{C_2} * 1)$$

$$= 10$$



H.W. Q:- Let $A = \{1, 2, 3, 4, 5\}$
How many partial order relation are possible on set A, such that the cardinality of each partial order relation is exactly '7'

Required partial order relation will be of type = $\{(1,1), (2,2), (3,3), (4,4), (5,5), (\ ,\ ), (\ ,\ )\}$

These two order pairs must be chosen in such a way that anti-symmetric and transitivity property is satisfied

H.W. Q:- Required partial order relation will be of type = $\{(1,1), (2,2), (3,3), (4,4), (5,5), (\ ,\ ), (\ ,\ )\}$

⇒ Two non-diagonal order pairs may be defined using:

(i) Exactly two distinct elements ⇒ i.e. of the type $(a,b), (b,a)$ — Not allowed because it will dis-satisfy anti-symmetric property

(ii) Exactly three distinct elements ⇒

i.e.

- $(a,b), (b,c)$ — Not allowed, it will dis-satisfy transitivity
- ✓ $(a,b), (a,c)$
- ✓ $(b,a), (c,a)$

These two choices will satisfy Anti-symmetric property, as well as transitivity.

(iii) Exacty four elements ⇒ i.e. ✓ $(a,b), (c,d)$ — such order pairs are also allowed w.r.t anti symmetric & transitivity property.

H.W. 9 :- Required Partial order relation will be of type = $\{(1,1),(2,2),(3,3),(4,4),(5,5),(\ .\ ),(\ .\ )\}$

$A = \{1,2,3,4,5\}$

Possible Combinations w.r.t. order pairs of type i.e. using three distinct elements

(i) $\{(a,b),(a,c)\} = {}^5C_3 * {}^3C_1 * 1$ (or)

(ii) $\{(b,a),(c,a)\} = {}^5C_3 * {}^3C_1 * 1$

Or

$= (10 * 3 * 1) = 30$

(or)

$+ \quad (10 * 3 * 1) = 30$

Possible Combinations w.r.t. order pairs of type i.e. using four distinct elements

$\{(a,b),(c,d)\} = {}^5C_4 * \left(\frac{4!}{2!}\right) = \left(5 * \frac{24}{2}\right) = 60$

$+$

$= {}^5C_4 * \left(\frac{({}^4C_2 * 2) * ({}^2C_2 * 2)}{2!}\right)$

$\left(\frac{({}^5C_2 * 2) * ({}^3C_2 * 2)}{2!}\right)$

Total "120" Partial order Rel^n possible on set A with Exactly 7 order Pairs.



2 mins Summary

- Topic: Bounded lattice
- Topic: Complement of an element in a lattice
- Topic: Complemented lattice

Slide

THANK - YOU